Week 4 progressive hints

Use the hints in order. After each hint, return to your own work and continue until you reach a new, specific impasse. The final hint may give a nearly complete setup, but it does not give the final result.

W04-Q01-PA

1. When finding f_x, treat y, e^y and y in the logarithmic term as constants. When finding f_y, treat x as constant.
2. Put f_x above f_y in a 2×1 column, then substitute $x = 1$, $y = 0$, $e^0 = 1$ and $\log 1 = 0$.

W04-Q01-PB

1. Differentiate each first partial derivative from Part A with respect to both variables.
2. Arrange $f_{xx}, f_{xy}, f_{yx}, f_{yy}$ in a 2×2 matrix and then evaluate it.
3. Ask whether all second partial derivatives are continuous on $(0, \infty) \times \mathbb{R}$; this is the condition relevant to the equality of mixed partials.

W04-Q02-PA

1. The x-derivative of e^{z-x} includes the derivative of $z - x$.
2. Put the three first partial derivatives in input order x, y, z, then evaluate e^{z-x} at the stated point.

W04-Q02-PB

1. The first-order scalar change is $\nabla g(\mathbf{a})^{\mathsf{T}} \mathrm{d}\mathbf{x}$.
2. Write the dimensions as $(1 \times 3)(3 \times 1)$ before taking the dot product.
3. A differential is the linear part of a local change. Distinguish it from evaluating $g(\mathbf{a} + \mathrm{d}\mathbf{x}) - g(\mathbf{a})$ exactly.

W04-Q03-PA

1. Find f_{xx}, f_{xy}, f_{yx} and f_{yy}, then evaluate them at $(1, -1)$.
2. First multiply the Hessian by $\mathbf{v}$, then left-multiply by $\mathbf{v}^{\mathsf{T}}$.
3. For the direct check, substitute $x = 1 + t$ and $y = -1 + 2t$ into f, expand only as far as needed, and differentiate the resulting polynomial twice.

W04-Q04-PA

1. Use $\nabla f(\mathbf{g}(t))^{\mathsf{T}} \mathbf{g}'(t)$, not a componentwise product.
2. At $t = 0$, evaluate the 2×1 gradient and the 2×1 path derivative separately before taking their inner product.

W04-Q04-PB

1. The inner Jacobian is 2×2, while the outer gradient at $\mathbf{G}(1, 0)$ is 2×1.
2. In column-gradient form,
$$\nabla h(1, 0) = \mathbf{J}_{\mathbf{G}}(1, 0)^{\mathsf{T}} \nabla F(\mathbf{G}(1, 0)).$$
3. For the direct check, use $\log(e^y) = y$ when writing $h(x, y)$, then differentiate the explicit expression.

W04-Q05-PA

1. Put the derivatives of F_1 in the first row and those of F_2 in the second row.
2. There are two outputs and three inputs, so use the required dimension as a check before substituting the point.

W04-Q05-PB

1. Multiply the 2×3 Jacobian from Part A by the 3×1 displacement.
2. Each row–column product gives the first-order change in the corresponding output component.
3. Keep the two changes in a 2×1 column and identify which entry belongs to F_1 and which belongs to F_2.

W04-Q06-PA

1. First calculate the intermediate point $\mathbf{g}(1, 2)$.
2. The inner Jacobian has one row per component of $\mathbf{g}$, while the outer Jacobian has one column per input of $\mathbf{F}$.
3. Multiply in the order
$$\mathbf{J}_{\mathbf{H}}(1, 2) = \mathbf{J}_{\mathbf{F}}(\mathbf{g}(1, 2))\mathbf{J}_{\mathbf{g}}(1, 2).$$

W04-Q06-PB

1. Substitute the three components of $\mathbf{g}(s, t)$ for x, y, z in each component of $\mathbf{F}$.
2. Differentiate each explicit component with respect to s and t; each output becomes one Jacobian row.
3. The Jacobian of a map from $\mathbb{R}^2$ to $\mathbb{R}^2$ must be 2×2. Compare this with the size of the reversed product.

W04-Q07-PA

1. Write $Y = 2K^{1/2}L^{1/2}$ and differentiate powers while holding the other input fixed.
2. Differentiate the first partials again to obtain the four Hessian entries. Evaluate exact powers such as $4^{1/2}$, $4^{-3/2}$ and $9^{-3/2}$.
3. For the change, use $\mathrm{d}Y = Y_K \, \mathrm{d}K + Y_L \, \mathrm{d}L$. Interpret each partial with the other input held fixed.

W04-Q08-PA

1. Differentiate the vector $\phi'(g)\nabla g$ component by component. One term differentiates $\phi'(g)$; the other differentiates ∇g.
2. The derivative of $\phi'(g)$ is $\phi''(g)\nabla g$. Its contribution is the outer product $\nabla g \nabla g^{\mathsf{T}}$.
3. For the application, take $g(x, y) = x^2 + 2y^2$ and $\phi(s) = \log(1 + s)$. Evaluate g, ∇g, $\mathbf{H}_g$, $\phi'(g)$ and $\phi''(g)$ at $(1, 1)$ before combining the two matrices.

W04-Q09-PA

1. Record $f(0)$, $f'(0)$ and $f''(0)$ for $f(x) = e^x$.
2. Insert these values into $T_2(x) = f(0) + f'(0)x + \frac{1}{2}f''(0)x^2$, then substitute $x = 0.1$.

W04-Q09-PB

1. The product has outer dimensions 2×1 because the inner dimensions 3 and 3 agree.
2. Take the inner product of each row of $\mathbf{A}$ with $\mathbf{b}$, preserving the row order.

W04-Q10-PA

1. Under the rate parameterisation, an exponential random variable has support $x \geq 0$ and survival probability $\mathbb{P}(X > x) = e^{-\lambda x}$.
2. Express the interval probability as $\mathbb{P}(X > 1/2) - \mathbb{P}(X > 1)$.

W04-Q10-PB

1. Expectation is linear: $\mathbb{E}[2X - Y] = 2\mathbb{E}[X] - \mathbb{E}[Y]$.

2. Use

$$\operatorname{Var}(aX + bY) = a^2 \operatorname{Var}(X) + b^2 \operatorname{Var}(Y) + 2ab \operatorname{Cov}(X, Y).$$

Here $a = 2$ and $b = -1$, so inspect the sign of $2ab$.